Railway planner - batch simulation report

Quantum Algorithm (Real Hardware) | 3 simulations | generated 11/7/2026, 10:37:28 PM

Setup

Algorithm	Quantum Algorithm (Real Hardware)
Budget	12
Nodes	4: Focsani, Braila, Galati, Buzau
Candidate routes	4: Buzau -> Braila (cost 2, value 7); Buzau -> Focsani (cost 2, value 6); Braila -> Galati (cost 1, value 22); Focsani -> Galati (cost 2, value 8)
Started	11/7/2026, 10:34:41 PM
Finished	11/7/2026, 10:37:08 PM

Overall statistics

Metric	Mean +/- std	Median	Min	Max	n
Approximation ratio (best sample / optimum)	1 +/- 0	1	1	1	3
Total value (valid plans)	43 +/- 0	43	43	43	3
Total cost (valid plans)	7 +/- 0	7	7	7	3
Valid sample rate	100.0% +/- 0.0%	100.0%	100.0%	100.0%	3
Connected sample rate	72.5% +/- 12.6%	78.3%	58.1%	81.2%	3
P(optimum) of the QAOA state	41.0% +/- 20.2%	49.7%	17.9%	55.5%	3
Planner runtime (s)	47.59 +/- 4.16	45.47	44.92	52.38	3
Request time (s)	48.68 +/- 4.07	46.65	46.03	53.37	3

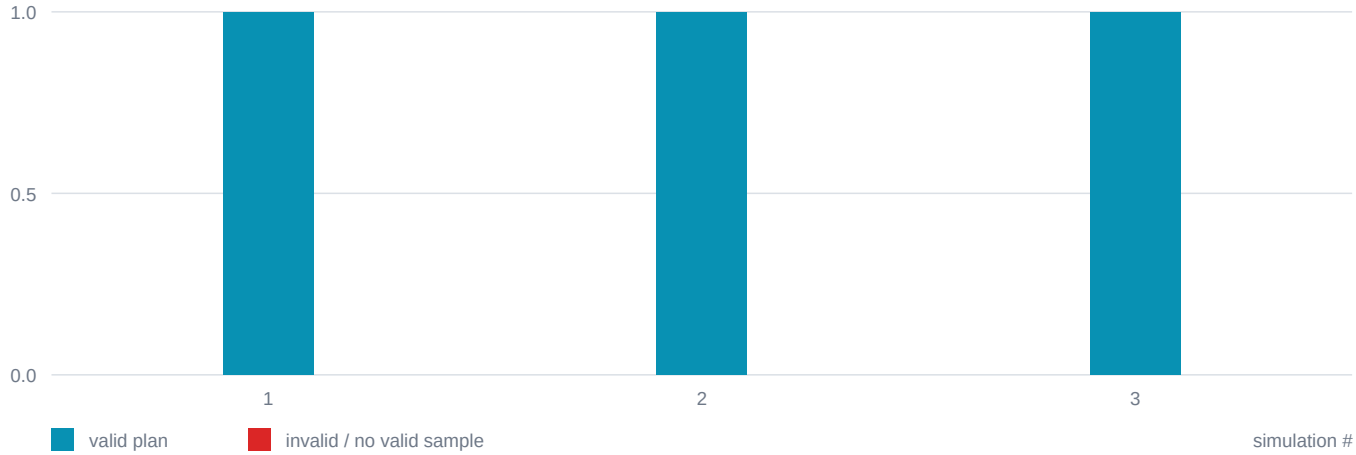
Finished 3 of 3 simulations (0 failed).

Valid plan rate: 100.0% | exact optimum found: 100.0%.

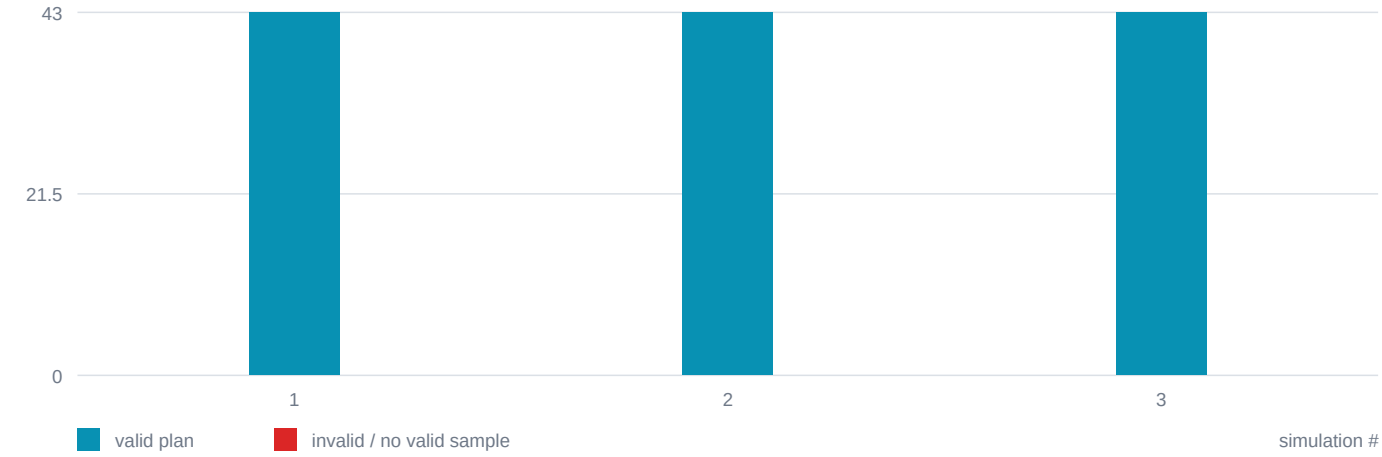
Best plan: simulation #1 (p=1 | slack | 300 it | 2048 shots, seed 2026) - value 43, cost 7.

Charts

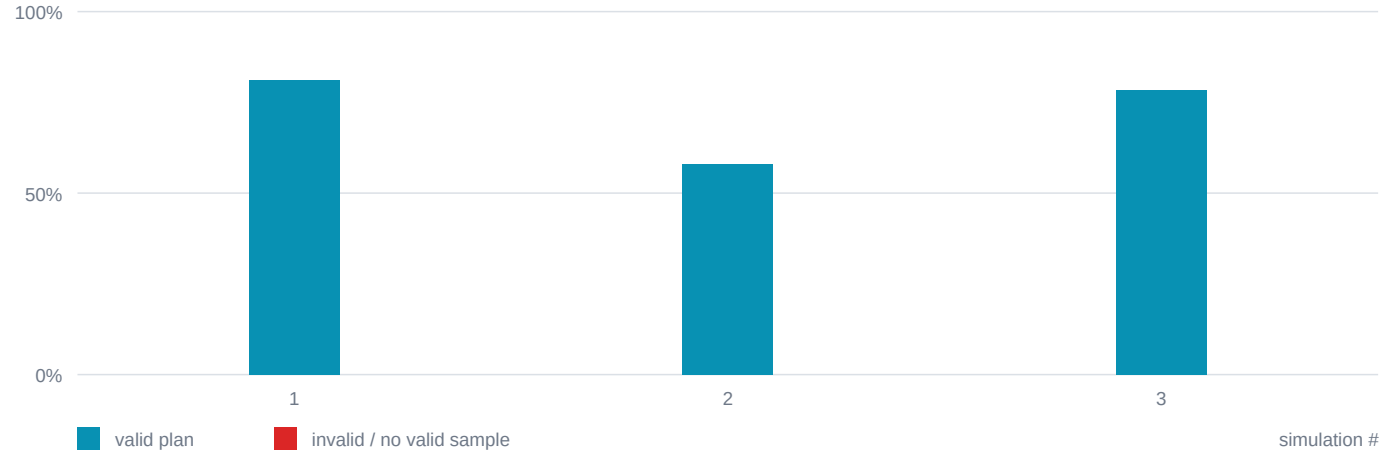
Approximation ratio per simulation



Total value per simulation



Connected sample rate per simulation



Statistics per parameter set

p	Slack	COBYLA it.	Shots	Runs	Valid	Optimum	Approx. ratio	Connected rate	Runtime (s)
1	slack	300	2048	1	100.0%	100.0%	1	81.2%	52.38
1	no slack	400	2048	1	100.0%	100.0%	1	58.1%	44.92
2	slack	200	800	1	100.0%	100.0%	1	78.3%	45.47

All simulations

#	p	Slack	It.	Shots	Seed	Status	Cost	Value	Optimum	Ratio	Valid rate	Conn. rate	Qubits	Time (s)
1	1	yes	300	2048	2026	valid	7	43	43	1	100.0%	81.2%	8	52.38
2	1	no	400	2048	2027	valid	7	43	43	1	100.0%	58.1%	4	44.92
3	2	yes	200	800	2028	valid	7	43	43	1	100.0%	78.3%	8	45.47

Circuits and QUBOs of the batch

#	p	Slack	Qubits	Depth	2q gates	Gate counts	QUBO lin./quad.	QUBO energy range	Expected energy
1	1	yes	8 (4 slack)	17	28	rzz 28, h 8, rz 8, rx 8, measure 8	8 / 28	-43 ... 37	-34.747
2	1	no	4	9	6	rzz 6, h 4, rz 4, rx 4, measure 4	4 / 6	-57.75 ... -33.56	-56.065
3	2	yes	8 (4 slack)	27	56	rzz 56, rz 16, rx 16, h 8, measure 8	8 / 28	-43 ... 37	-37.297

Simulation #1

p = 1 | slack | 300 COBYLA iterations | 2048 shots | seed 2026 | valid plan

Intermediate values

#	Step	Value
1	**Planner result**	Connected network returned within the 12 budget.
2	**QAOA configuration**	ibm-qpu | 8 qubits | 4 route + 4 slack | p = 1 | slack bits | 300 COBYLA iterations | 2048 shots
3	**Penalty weights**	slack bits | lambda 0.25 | mu (conflicts) 44 | beta (city cover) 0.25 | slack weights 1, 2, 4, 5 | budget 12 units
4	**Optimised angles**	gamma [-0.862] | beta [0.6396]
5	**Angle training (COBYLA)**	7 optimiser runs, 613 energy evaluations | warm start auto | selected previous / ramp dt=0.02
6	**Optimiser run 1**	interp | p = 1 | start grid scan | E -34.698 -> -34.747 | 382 evaluations | gamma [-0.862] | beta [0.6396]
7	**Optimiser run 2 (selected)**	previous | p = 1 | start ramp dt=0.02 | E -16.43 -> -34.747 | 38 evaluations | gamma [-0.862] | beta [0.6396]
8	**Optimiser run 3**	previous | p = 1 | start ramp dt=0.05 | E -18.464 -> -34.747 | 36 evaluations | gamma [-0.862] | beta [0.6395]
9	**Optimiser run 4**	previous | p = 1 | start ramp dt=0.1 | E -23.206 -> -34.747 | 39 evaluations | gamma [-0.8619] | beta [0.6396]
10	**Optimiser run 5**	previous | p = 1 | start random | E -21.844 -> -34.747 | 36 evaluations | gamma [-0.8619] | beta [0.6395]
11	**Optimiser run 6**	previous | p = 1 | start random | E -16.745 -> -34.747 | 40 evaluations | gamma [-0.8618] | beta [0.6395]
12	**Optimiser run 7**	previous | p = 1 | start random | E -17.743 -> -34.747 | 42 evaluations | gamma [-0.8619] | beta [0.6395]
13	**QAOA circuit**	8 qubits | depth 17 | 28 two-qubit gates
14	**Expected energy**	-34.747
15	**P(valid) of the QAOA state**	100.00% (uniform 100.00%)
16	**P(connected) of the QAOA state**	88.70%
17	**P(optimum) of the QAOA state**	49.66%
18	**Valid sample rate**	100.00%
19	**Connected sample rate**	81.15%
20	**Best sampled value**	43
21	**Exact optimum value**	43
22	**Approximation ratio**	1 | exact optimum found
23	**Reference optimum (exhaustive check)**	value 43 | cost 7 | Buzau -> Braila, Buzau -> Focsani, Braila -> Galati, Focsani -> Galati
24	**Top sample 1**	848 shots | value 43 | cost 7 | valid, connected | Buzau -> Braila, Buzau -> Focsani, Braila -> Galati, Focsani -> Galati
25	**Top sample 2**	286 shots | value 37 | cost 5 | valid, connected | Buzau -> Braila, Braila -> Galati, Focsani -> Galati
26	**Top sample 3**	254 shots | value 36 | cost 5 | valid, connected | Buzau -> Focsani, Braila -> Galati, Focsani -> Galati
27	**Top sample 4**	183 shots | value 35 | cost 5 | valid, connected | Buzau -> Braila, Buzau -> Focsani, Braila -> Galati
28	**Top sample 5**	91 shots | value 21 | cost 6 | valid, connected | Buzau -> Braila, Buzau -> Focsani, Focsani -> Galati
29	**Top sample 6**	88 shots | value 30 | cost 3 | valid, not connected | Braila -> Galati, Focsani -> Galati
30	**Top sample 7**	86 shots | value 29 | cost 3 | valid, not connected | Buzau -> Braila, Braila -> Galati
31	**Top sample 8**	67 shots | value 28 | cost 3 | valid, not connected | Buzau -> Focsani, Braila -> Galati
32	**Planner timing**	penalty tuning 0.14 s | angle training 0.3 s | sampling 51.908 s
33	**Hardware execution**	ibm_aachen | status done | 2048 shots | job dbnomq24udtc73c3v3hg | auto layout | compiled depth 174 | 89 two-qubit gates | estimated success 82.91%
34	**Cost breakdown**	Quantum 7 | connectors 0 | extra 0
35	**quantum route**	Buzau -> Braila | cost 2 | reward +7
36	**quantum route**	Buzau -> Focsani | cost 2 | reward +6
37	**quantum route**	Braila -> Galati | cost 1 | reward +22
38	**quantum route**	Focsani -> Galati | cost 2 | reward +8

QUBO

Budget encoding	slack bits
Offset	37
Energy range	-43 ... 37
Terms	8 linear, 28 quadratic
Variables	x0 = Buzau -> Braila; x1 = Buzau -> Focsani; x2 = Braila -> Galati; x3 = Focsani -> Galati; s0 = slack 1; s1 = slack 2; s2 = slack 4; s3 = slack 5

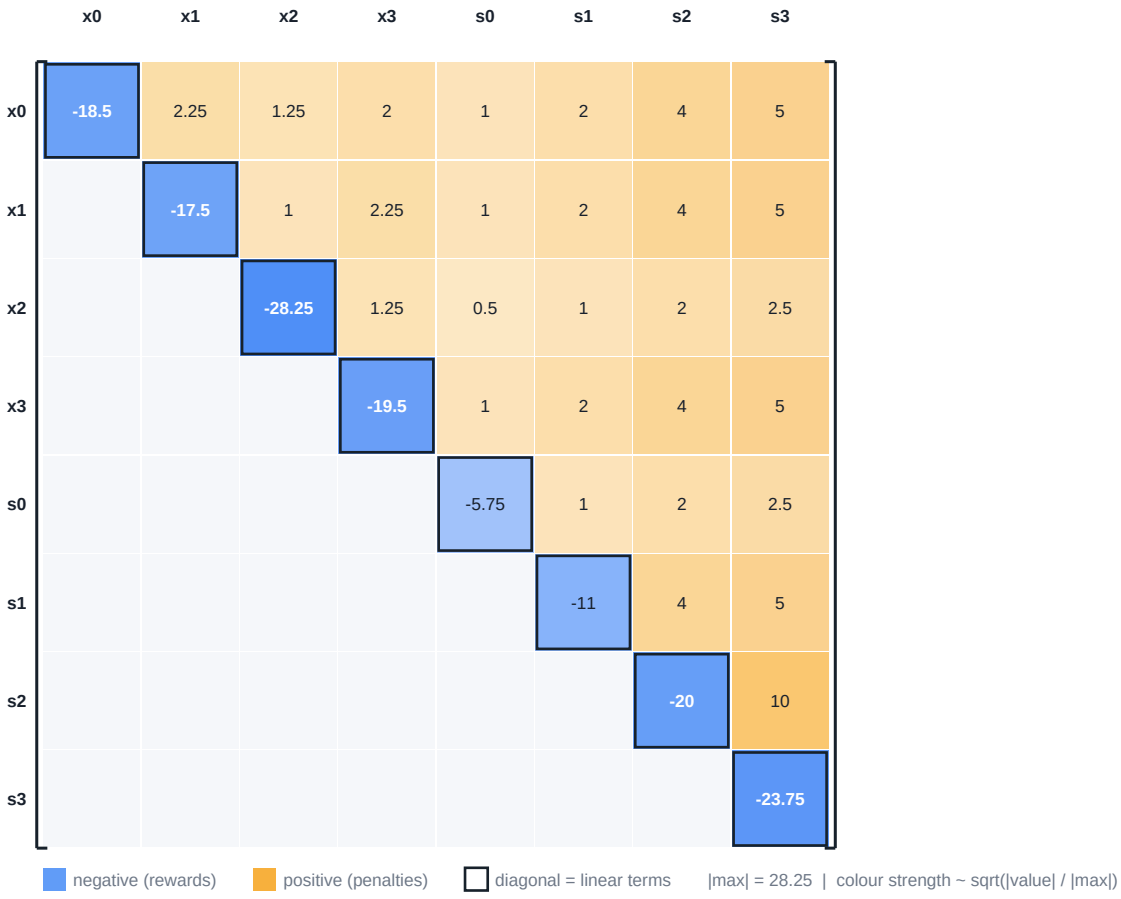
Formula

```
E = -(7 x0 + 6 x1 + 22 x2 + 8 x3) + 0.25(2 x0 + 2 x1 + x2 + 2 x3 + s0 + 2 s1 + 4 s2 + 5 s3 - 12)^2 + 0.25((1 - x2 - x3 +
1*(x2*x3)) + (1 - x0 - x2 + 1*(x0*x2)) + (1 - x0 - x1 + 1*(x0*x1)) + (1 - x1 - x3 + 1*(x1*x3)))
```

Term	Coefficient	Term	Coefficient	Term	Coefficient
x0	-18.5	x1	-17.5	x2	-28.25
x3	-19.5	s0	-5.75	s1	-11
s2	-20	s3	-23.75	x0*x1	2.25
x0*x2	1.25	x0*x3	2	x0*s0	1
x0*s1	2	x0*s2	4	x0*s3	5
x1*x2	1	x1*x3	2.25	x1*s0	1
x1*s1	2	x1*s2	4	x1*s3	5
x2*x3	1.25	x2*s0	0.5	x2*s1	1
x2*s2	2	x2*s3	2.5	x3*s0	1
x3*s1	2	x3*s2	4	x3*s3	5
s0*s1	1	s0*s2	2	s0*s3	2.5
s1*s2	4	s1*s3	5	s2*s3	10

QUBO matrix (8 x 8)

upper-triangular Q: row/column order = matrix_variables, diagonal Q[i][i] = linear coefficient of y_i, Q[i][j] (i < j) = coefficient of y_i*y_j, below the diagonal 0; E(y) = offset + y^T Q y (bits, so y_i^2 = y_i)



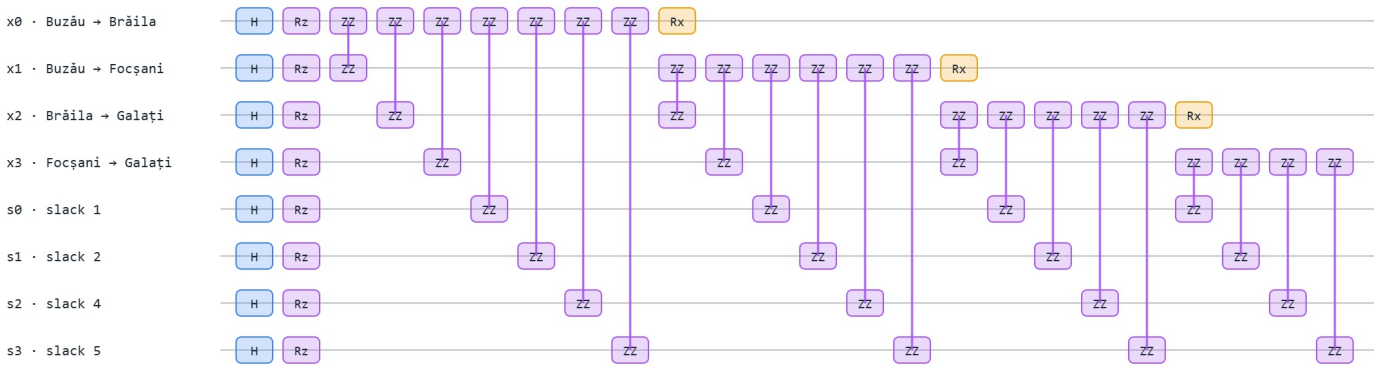
QAOA circuit

8 qubits | depth 17 | 28 two-qubit gates | rzz 28, h 8, rz 8, rx 8, measure 8

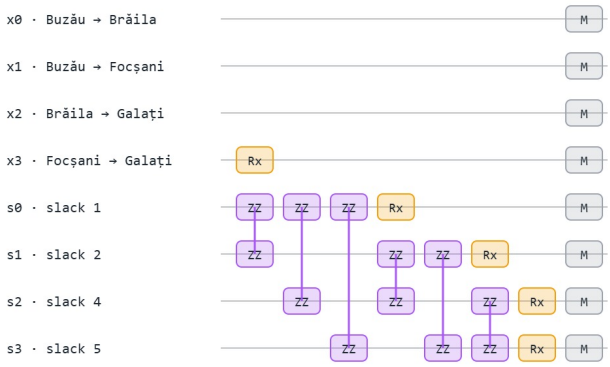
gamma = [-0.8620] beta = [0.6396]

Hadamard Rz / Rzz (cost, gamma) Rx (mixer, beta) Measure

gate columns 1-24 of 32



gate columns 25-32 of 32



Simulation #2

p = 1 | no slack | 400 COBYLA iterations | 2048 shots | seed 2027 | valid plan

Intermediate values

#	Step	Value
1	**Planner result**	Connected network returned within the 12 budget.
2	**QAOA configuration**	ibm-qpu | 4 qubits | 4 route + 0 slack | p = 1 | unbalanced penalization (no slack) | 400 COBYLA iterations | 2048 shots
3	**Penalty weights**	unbalanced penalization (no slack) | lambda1 3 | lambda2 0.01 | mu (conflicts) 44 | beta (city cover) 0.25 | budget 12 units
4	**Optimised angles**	gamma [-0.7969] | beta [0.7851]
5	**Angle training (COBYLA)**	7 optimiser runs, 593 energy evaluations | warm start auto | selected interp / grid scan
6	**Optimiser run 1 (selected)**	interp | p = 1 | start grid scan | E -56.065 -> -56.065 | 378 evaluations | gamma [-0.7969] | beta [0.7851]
7	**Optimiser run 2**	previous | p = 1 | start ramp dt=0.02 | E -46.184 -> -56.065 | 36 evaluations | gamma [-0.7969] | beta [0.7851]
8	**Optimiser run 3**	previous | p = 1 | start ramp dt=0.05 | E -47.112 -> -56.065 | 35 evaluations | gamma [-0.7969] | beta [0.785]
9	**Optimiser run 4**	previous | p = 1 | start ramp dt=0.1 | E -49.61 -> -56.065 | 31 evaluations | gamma [-0.7969] | beta [0.7852]
10	**Optimiser run 5**	previous | p = 1 | start random | E -51.189 -> -56.065 | 35 evaluations | gamma [-0.797] | beta [0.785]
11	**Optimiser run 6**	previous | p = 1 | start random | E -50.778 -> -56.065 | 42 evaluations | gamma [-0.7969] | beta [0.7851]
12	**Optimiser run 7**	previous | p = 1 | start random | E -46.484 -> -56.065 | 36 evaluations | gamma [-0.7968] | beta [0.785]
13	**QAOA circuit**	4 qubits | depth 9 | 6 two-qubit gates
14	**Expected energy**	-56.065
15	**P(valid) of the QAOA state**	100.00% (uniform 100.00%)
16	**P(connected) of the QAOA state**	59.69%
17	**P(optimum) of the QAOA state**	17.93%
18	**Valid sample rate**	100.00%
19	**Connected sample rate**	58.06%
20	**Best sampled value**	43
21	**Exact optimum value**	43
22	**Approximation ratio**	1 | exact optimum found
23	**Reference optimum (exhaustive check)**	value 43 | cost 7 | Buzau -> Braila, Buzau -> Focsani, Braila -> Galati, Focsani -> Galati
24	**Top sample 1**	352 shots | value 37 | cost 5 | valid, connected | Buzau -> Braila, Braila -> Galati, Focsani -> Galati
25	**Top sample 2**	328 shots | value 43 | cost 7 | valid, connected | Buzau -> Braila, Buzau -> Focsani, Braila -> Galati, Focsani -> Galati
26	**Top sample 3**	261 shots | value 36 | cost 5 | valid, connected | Buzau -> Focsani, Braila -> Galati, Focsani -> Galati
27	**Top sample 4**	249 shots | value 30 | cost 3 | valid, not connected | Braila -> Galati, Focsani -> Galati
28	**Top sample 5**	235 shots | value 35 | cost 5 | valid, connected | Buzau -> Braila, Buzau -> Focsani, Braila -> Galati
29	**Top sample 6**	214 shots | value 29 | cost 3 | valid, not connected | Buzau -> Braila, Braila -> Galati
30	**Top sample 7**	186 shots | value 28 | cost 3 | valid, not connected | Buzau -> Focsani, Braila -> Galati
31	**Top sample 8**	174 shots | value 22 | cost 1 | valid, not connected | Braila -> Galati
32	**Planner timing**	penalty tuning 1.156 s | angle training 0.173 s | sampling 43.582 s
33	**Hardware execution**	ibm_aachen | status done | 2048 shots | job dbnon7g0rnvc73973b90 | auto layout | compiled depth 63 | 19 two-qubit gates | estimated success 96.66%
34	**Cost breakdown**	Quantum 7 | connectors 0 | extra 0
35	**quantum route**	Buzau -> Braila | cost 2 | reward +7
36	**quantum route**	Buzau -> Focsani | cost 2 | reward +6
37	**quantum route**	Braila -> Galati | cost 1 | reward +22
38	**quantum route**	Focsani -> Galati | cost 2 | reward +8

QUBO

Budget encoding	unbalanced penalization (no slack)
Offset	-33.56
Energy range	-57.75 ... -33.56
Terms	4 linear, 6 quadratic
Variables	x0 = Buzau -> Braila; x1 = Buzau -> Focsani; x2 = Braila -> Galati; x3 = Focsani -> Galati

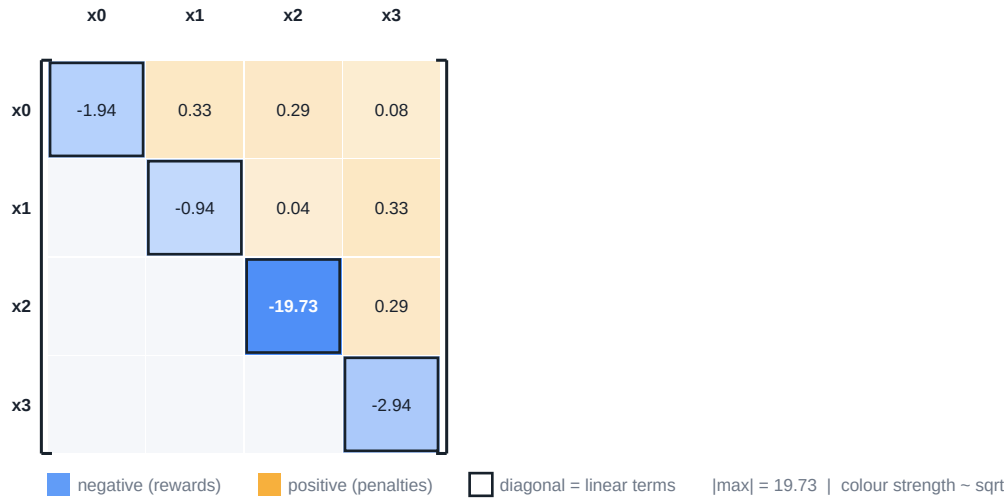
Formula

```
E = -(7 x0 + 6 x1 + 22 x2 + 8 x3) - 3*(12 - (2 x0 + 2 x1 + x2 + 2 x3)) + 0.01*(12 - (2 x0 + 2 x1 + x2 + 2 x3))^2 + 0.25((1 - x2 - x3 + 1*(x2*x3)) + (1 - x0 - x2 + 1*(x0*x2)) + (1 - x0 - x1 + 1*(x0*x1)) + (1 - x1 - x3 + 1*(x1*x3)))
```

Term	Coefficient	Term	Coefficient	Term	Coefficient
x0	-1.94	x1	-0.94	x2	-19.73
x3	-2.94	x0*x1	0.33	x0*x2	0.29
x0*x3	0.08	x1*x2	0.04	x1*x3	0.33
x2*x3	0.29				

QUBO matrix (4 x 4)

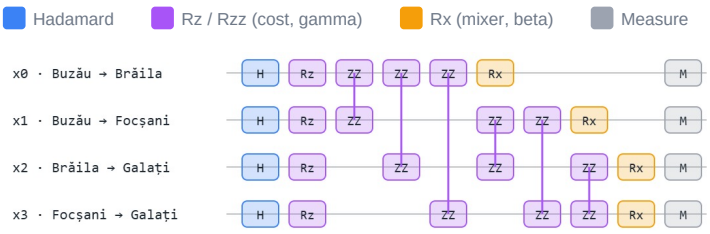
upper-triangular Q: row/column order = matrix_variables, diagonal Q[i][i] = linear coefficient of y_i, Q[i][j] (i < j) = coefficient of y_i*y_j, below the diagonal 0; E(y) = offset + y^T Q y (bits, so y_i^2 = y_i)



QAOA circuit

4 qubits | depth 9 | 6 two-qubit gates | rzz 6, h 4, rz 4, rx 4, measure 4

gamma = [-0.7969] beta = [0.7851]



Simulation #3

p = 2 | slack | 200 COBYLA iterations | 800 shots | seed 2028 | valid plan

Intermediate values

#	Step	Value
1	**Planner result**	Connected network returned within the 12 budget.
2	**QAOA configuration**	ibm-qpu | 8 qubits | 4 route + 4 slack | p = 2 | slack bits | 200 COBYLA iterations | 800 shots
3	**Penalty weights**	slack bits | lambda 0.25 | mu (conflicts) 44 | beta (city cover) 0.25 | slack weights 1, 2, 4, 5 | budget 12 units
4	**Optimised angles**	gamma [-0.6444, -1.4396] | beta [0.9518, 0.3407]
5	**Angle training (COBYLA)**	8 optimiser runs, 1782 energy evaluations | warm start auto | selected previous / ramp dt=0.1
6	**Optimiser run 1**	interp | p = 1 | start grid scan | E -34.698 -> -34.747 | 382 evaluations | gamma [-0.862] | beta [0.6396]
7	**Optimiser run 2**	interp | p = 2 | start INTERP | E -26.325 -> -37.296 | 200 evaluations | gamma [-0.6397, -1.4338] | beta [0.9498, 0.3417]
8	**Optimiser run 3**	previous | p = 2 | start ramp dt=0.02 | E -16.931 -> -36.819 | 200 evaluations | gamma [-0.4421, -1.0168] | beta [0.8244, 0.3992]
9	**Optimiser run 4**	previous | p = 2 | start ramp dt=0.05 | E -20.686 -> -37.277 | 200 evaluations | gamma [-0.6116, -1.3801] | beta [0.948, 0.3495]
10	**Optimiser run 5 (selected)**	previous | p = 2 | start ramp dt=0.1 | E -26.059 -> -37.297 | 200 evaluations | gamma [-0.6444, -1.4396] | beta [0.9518, 0.3407]
11	**Optimiser run 6**	previous | p = 2 | start random | E -13.65 -> -34.757 | 200 evaluations | gamma [0.8789, 0.0622] | beta [1.0556, 1.4467]
12	**Optimiser run 7**	previous | p = 2 | start random | E -23.61 -> -37.143 | 200 evaluations | gamma [0.5347, -0.8629] | beta [2.2974, 0.3251]
13	**Optimiser run 8**	previous | p = 2 | start random | E -25.491 -> -37.28 | 200 evaluations | gamma [-0.6151, -1.3885] | beta [0.95, 0.3481]
14	**QAOA circuit**	8 qubits | depth 27 | 56 two-qubit gates
15	**Expected energy**	-37.297
16	**P(valid) of the QAOA state**	100.00% (uniform 100.00%)
17	**P(connected) of the QAOA state**	93.31%
18	**P(optimum) of the QAOA state**	55.53%
19	**Valid sample rate**	100.00%
20	**Connected sample rate**	78.25%
21	**Best sampled value**	43
22	**Exact optimum value**	43
23	**Approximation ratio**	1 | exact optimum found
24	**Reference optimum (exhaustive check)**	value 43 | cost 7 | Buzau -> Braila, Buzau -> Focsani, Braila -> Galati, Focsani -> Galati
25	**Top sample 1**	303 shots | value 43 | cost 7 | valid, connected | Buzau -> Braila, Buzau -> Focsani, Braila -> Galati, Focsani -> Galati
26	**Top sample 2**	112 shots | value 37 | cost 5 | valid, connected | Buzau -> Braila, Braila -> Galati, Focsani -> Galati
27	**Top sample 3**	94 shots | value 35 | cost 5 | valid, connected | Buzau -> Braila, Buzau -> Focsani, Braila -> Galati
28	**Top sample 4**	91 shots | value 36 | cost 5 | valid, connected | Buzau -> Focsani, Braila -> Galati, Focsani -> Galati
29	**Top sample 5**	39 shots | value 29 | cost 3 | valid, not connected | Buzau -> Braila, Braila -> Galati
30	**Top sample 6**	38 shots | value 30 | cost 3 | valid, not connected | Braila -> Galati, Focsani -> Galati
31	**Top sample 7**	27 shots | value 28 | cost 3 | valid, not connected | Buzau -> Focsani, Braila -> Galati
32	**Top sample 8**	26 shots | value 21 | cost 6 | valid, connected | Buzau -> Braila, Buzau -> Focsani, Focsani -> Galati
33	**Planner timing**	penalty tuning 0.148 s | angle training 1.899 s | sampling 43.372 s
34	**Hardware execution**	ibm_aachen | status done | 800 shots | job dbnonjift1mc73fl3mu0 | auto layout | compiled depth 406 | 209 two-qubit gates | estimated success 70.10%
35	**Cost breakdown**	Quantum 7 | connectors 0 | extra 0
36	**quantum route**	Buzau -> Braila | cost 2 | reward +7
37	**quantum route**	Buzau -> Focsani | cost 2 | reward +6
38	**quantum route**	Braila -> Galati | cost 1 | reward +22
39	**quantum route**	Focsani -> Galati | cost 2 | reward +8

QUBO

Budget encoding	slack bits
Offset	37
Energy range	-43 ... 37
Terms	8 linear, 28 quadratic
Variables	x0 = Buzau -> Braila; x1 = Buzau -> Focsani; x2 = Braila -> Galati; x3 = Focsani -> Galati; s0 = slack 1; s1 = slack 2; s2 = slack 4; s3 = slack 5

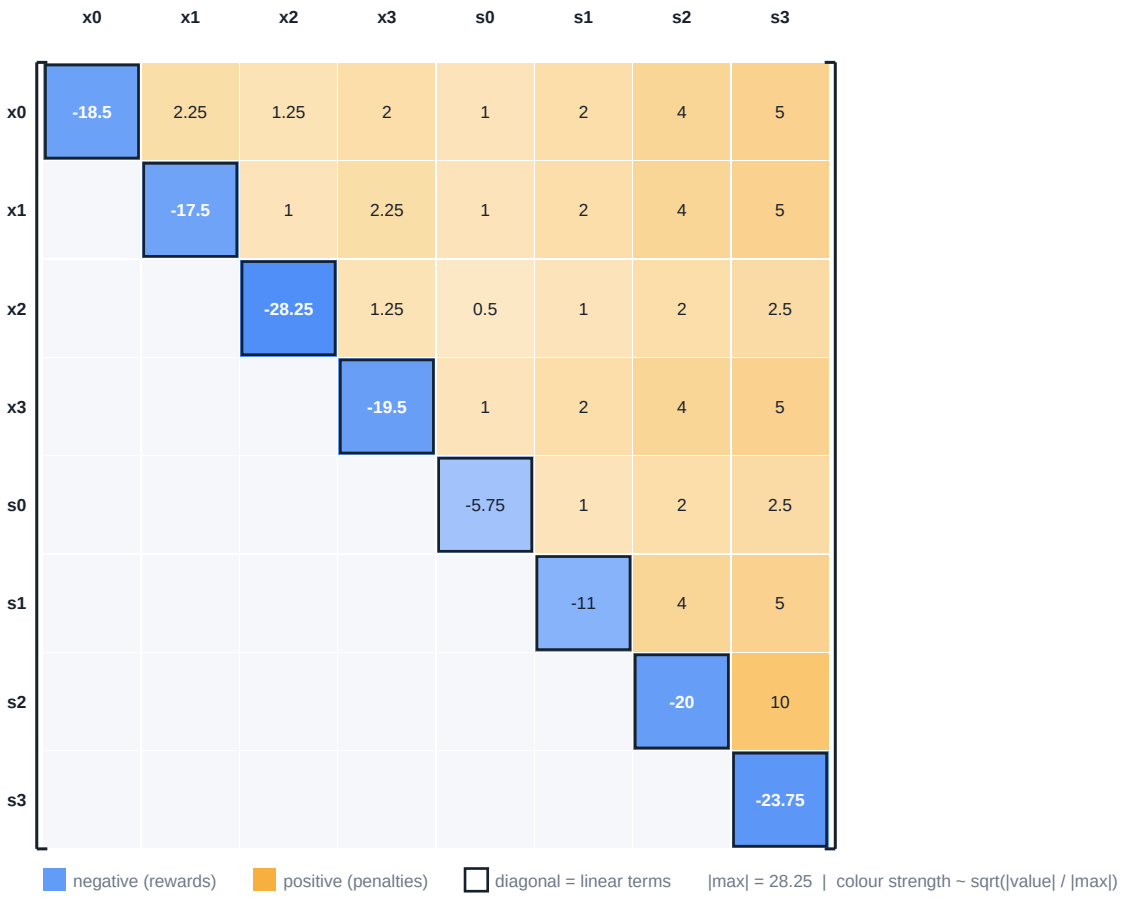
Formula

```
E = -(7 x0 + 6 x1 + 22 x2 + 8 x3) + 0.25(2 x0 + 2 x1 + x2 + 2 x3 + s0 + 2 s1 + 4 s2 + 5 s3 - 12)^2 + 0.25((1 - x2 - x3 + 1*(x2*x3)) + (1 - x0 - x2 + 1*(x0*x2)) + (1 - x0 - x1 + 1*(x0*x1)) + (1 - x1 - x3 + 1*(x1*x3)))
```

Term	Coefficient	Term	Coefficient	Term	Coefficient
x0	-18.5	x1	-17.5	x2	-28.25
x3	-19.5	s0	-5.75	s1	-11
s2	-20	s3	-23.75	x0*x1	2.25
x0*x2	1.25	x0*x3	2	x0*s0	1
x0*s1	2	x0*s2	4	x0*s3	5
x1*x2	1	x1*x3	2.25	x1*s0	1
x1*s1	2	x1*s2	4	x1*s3	5
x2*x3	1.25	x2*s0	0.5	x2*s1	1
x2*s2	2	x2*s3	2.5	x3*s0	1
x3*s1	2	x3*s2	4	x3*s3	5
s0*s1	1	s0*s2	2	s0*s3	2.5
s1*s2	4	s1*s3	5	s2*s3	10

QUBO matrix (8 x 8)

upper-triangular Q: row/column order = matrix_variables, diagonal Q[i][i] = linear coefficient of y_i, Q[i][j] (i < j) = coefficient of y_i*y_j, below the diagonal 0; E(y) = offset + y^T Q y (bits, so y_i^2 = y_i)



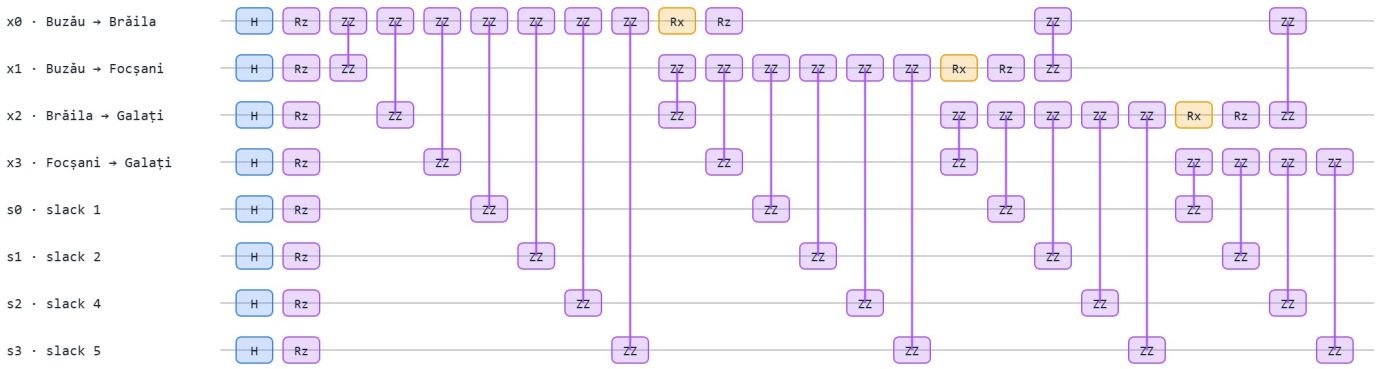
QAOA circuit

8 qubits | depth 27 | 56 two-qubit gates | rzz 56, rz 16, rx 16, h 8, measure 8

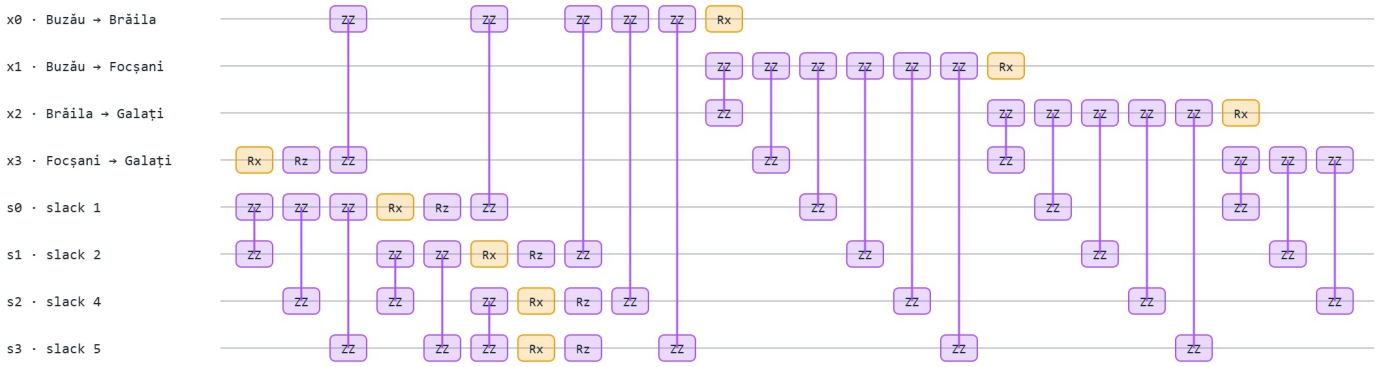
gamma = [-0.6444, -1.4396] beta = [0.9518, 0.3407]

Hadamard Rz / Rzz (cost, gamma) Rx (mixer, beta) Measure

gate columns 1-24 of 57



gate columns 25-48 of 57



gate columns 49-57 of 57

